



Discriminative kernel convolution network for multi-label ophthalmic disease detection on imbalanced fundus image dataset

Amit Bhati^a, Neha Gour^{a,1}, Pritee Khanna^{a,*}, Aparajita Ojha^a

^a Department of Computer Science and Engineering, PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur 482005, India

ARTICLE INFO

Keywords:

Multi-label classification
Channel shuffle
Discriminative kernel convolution (DKCNet)
Fundus image
ODIR-5K

ABSTRACT

It is feasible to recognize the presence and seriousness of eye disease by investigating the progressions in retinal biological structures. Fundus examination is a diagnostic procedure to examine the biological structure and anomalies present in the eye. Ophthalmic diseases like glaucoma, diabetic retinopathy, and cataracts are the main cause of visual impairment worldwide. Ocular Disease Intelligent Recognition (ODIR-5K) is a benchmark structured fundus image dataset utilized by researchers for multi-label multi-disease classification of fundus images. This work presents a Discriminative Kernel Convolution Network (DKCNet), which explores discriminative region-wise features without adding extra computational cost. DKCNet is composed of an attention block followed by a Squeeze-and-Excitation (SE) block. The attention block takes features from the backbone network and generates discriminative feature attention maps. The SE block takes the discriminative feature maps and improves channel interdependencies. Better performance of DKCNet is observed with InceptionResnet backbone network for multi-label classification of ODIR-5K fundus images with 96.08 AUC, 94.28 F1-score, and 0.81 kappa score. The proposed method splits the common target label for an eye pair based on the diagnostic keyword. Based on these labels, over-sampling and/or under-sampling are done to resolve the class imbalance. To check the bias of the proposed model towards training data, the model trained on the ODIR dataset is tested on three publicly available benchmark datasets. It is observed that the proposed DKCNet gives good performance on completely unseen fundus images also.

1. Introduction

Ophthalmic diseases are the leading cause of blindness worldwide. A report published by the World Health Organization (WHO) in 2021 says that around 2.2 billion people are visually impaired, and almost half of these cases are preventable based on timely detection and treatment [1]. The human retina is a light-sensitive layer of tissues in the rear end of the eye. The incident light is converted into neural signals through the receptors on the retina and handled by the brain's visual cortex to generate an image. The retina gets affected by different abnormalities which influences human vision [2]. Fundus, fluorescein angiography, and Optical Coherence Tomography (OCT) are standard modalities used by experts to investigate ophthalmic diseases [3]. Fundus imaging is the primary image modality utilized for the clinical examination of ophthalmic diseases.

The presence of ophthalmic diseases can be recognized by observing abnormalities close to retinal areas like optic nerve, veins, macula, optic plate, etc. Early identification of retinal abnormalities depicted in Fig. 1 is crucial, yet difficult, as a few signs appear in

the beginning stage. Deep neural networks are effectively utilized for retinal vessel segmentation, lesion detection [4–6], and glaucoma or diabetic retinopathy stage classification [7,8]. Traditional single-label classification, also known as multi-class classification, includes a single class label for each instance. However, ophthalmic disease classification is a complex problem as multiple labels may be associated with a single instance. Among publicly available fundus image datasets, e.g., Messidor, EyePACS, IDRiD, STARE, DRIVE, Drishti-GS1, Ocular Disease Intelligent Recognition (ODIR-5K) etc. [9], only ODIR-5K dataset [10] presents the real-life challenge of multi-class multi-label ophthalmic disease detection. The other datasets are specific to a single disease only. However, ODIR-5K is an imbalanced dataset. The bias towards the majority class in an imbalanced dataset affects the training and classification accuracy of models. Further, every disease has various phases of severity that can be verified by visualizing the existence and shape of various lesions. Although lesions are characterized by specific morphological features, sometimes lesions of different diseases may show similar characteristics, which makes the disease detection task more complex.

* Corresponding author.

E-mail address: pkhanna@iiitdmj.ac.in (P. Khanna).

¹ Permanent Address: KUCARS and C2PS, Departement of Electrical Engineering and Computer Science, Khalifa University, Abu Dhabi, United Arab Emirates.

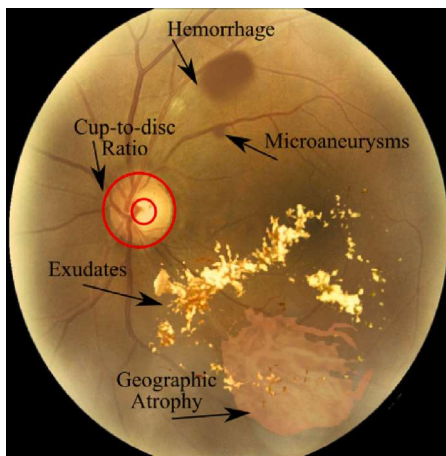


Fig. 1. Fundus image of an eye with different kinds of abnormalities [11].

This work presents a deep neural network-based framework for multi-label ophthalmic disease classification of the fundus image. A patient might be impacted by several eye diseases at the same time, which makes this classification problem a multi-class, multi-label problem. The available state-of-the-art methods take left and right eye fundus images simultaneously and predict classification labels of diseases present in the eye pairs. But sometimes it would be the case when one eye may have several diseases and the other eye may be normal, hence these models could not identify which eye suffered from which kind of disease. In the ODIR dataset, the disease diagnostic keywords were assigned to each image by eye specialists. These diagnostic keywords contain disease symptomatic details by which labels are decided. For example, the left eye diagnostic keywords for patient number 188 are laser spot and severe proliferative diabetic retinopathy, whereas the right eye diagnostic keyword is cataract and the common labels are D, C and O. Since the proposed method splits the common disease labels of eye pairs into independent labels for left and right eye disease based on diagnosis keywords, the model is trained for individual eye images with their corresponding labels to diagnose the presence of multiple diseases in a particular eye. It is also observed that class imbalance is a prevalent issue in medical image datasets, which affects the classification accuracy of models. The proposed architecture is designed to handle the issue of class imbalance in the fundus image dataset. A dilated convolution-based attention network named Discriminative Kernel Convolution Network (DKCNet) can simultaneously detect multiple lesion parts related to ophthalmic diseases appearing in a fundus image. The proposed DKCNet is observed to achieve better performance for multi-label ophthalmic disease classification as compared to the methods proposed in the literature.

The work is organized as follows. Section 2 connects retinal abnormalities present in fundus images with ophthalmic diseases. State-of-the-art techniques for fundus image classification are discussed in Section 3. Section 4 describes the proposed transfer learning-based DKCNet model for fundus image classification. The dataset used for experimental evaluation is also discussed here. The results are discussed in Section 5 and advantages, limitations, and future directions in Section 6.

2. Retinal abnormalities and ophthalmic diseases

Fig. 2 shows fundus images belonging to different disease classes from the ODIR-5K dataset. Glaucoma is an eye condition that affects the optic nerve, whose strength is crucial for good vision. This harm is typically caused by an unusually high pressure in the eye [12]. Glaucoma can be distinguished by noticing changes in the proportion of the

optic disc cup and neuro-retinal edge surface region known as the cup-to-disc ratio. Diabetic retinopathy is a diabetes intricacy that influences the tissues inside the retina. Diabetic retinopathy may not show any adverse indications, and patients may have minor issues related to clear vision. However, it may lead to visual impairment if not treated at an earlier stage. The macula is liable for clear central vision, and the distortion in vision starts if liquids collect in it. Age-related Macular Degeneration (AMD) can be distinguished by noticing the growth of fresh blood vessels or the presence of dead retinal cells known as neovascularization and geographic atrophy, respectively [13]. The causes of cataracts include the development of patches that make vision difficult [14]. The presence of cataracts can be observed when the optic disc, fovea, and other parts of the eye become hazy. Hypertension is a silent illness. This disease changes the biological shapes of veins, like length and thickness, and results in cardiovascular disease, stroke, and respiratory failures over the long run [15]. Myopia causes critical vision loss because of diminishing epithelial tissues and changes in eye colour. It fundamentally modifies the visualization of any object from a certain distance by making them blurred. The fundus images can also be used to detect some other anomalies like pigment epithelium proliferation, epiretinal membrane, tessellated fundus, and vitreous macular degeneration.

3. Related works

Deep learning-based ocular image analysis can be used in conjunction with telemedicine for diagnosing and controlling ophthalmic diseases at an early stage [16]. Many researchers have made significant efforts to resolve the multi-label classification problem of ophthalmic disease. All these works are simulated on the publicly available ODIR-5K dataset. Islam et al. [17] proposed a shallow Convolution Neural Network (CNN)-based model trained from scratch for the classification of fundus images of ODIR-5K dataset. The left and right eye fundus images are input to the CNN model independently, and the disease label is assigned accordingly. Although the model is less complex, it fails to distinguish multiple diseases. Wang et al. [18] preprocessed fundus images using grey and colour histogram equalization. Various data augmentation techniques are also used. The preprocessed images are applied to two parallel EfficientNet models, and feature concatenation is done at the last layer for classification. However, they could achieve only 73% AUC and 88% F1-Score on ODIR-5K dataset.

Li et al. [19] proposed a Dense Correlation Network (DCNet) using transfer learning-based ResNet architecture. The Spatial Correlation Module (SCM) is the basic building block of this network architecture. The SCM block defines pixel-wise dense correlation between features extracted from colour fundus images. These correlated features are fused to create the final feature map for ophthalmic disease classification in 8 classes of ODIR-5K dataset with 93% AUC and 91.3% F1-score. Similarly, Gour and Khanna [20] proposed a pre-trained, two-input CNN architecture for the ODIR-5K dataset. They passed left and right eye fundus images to two pre-trained VGG-16 networks [21] in parallel. The extracted features are concatenated to create the feature map. Despite using the VGG model, they failed to beat the performance of [19].

Li et al. [22] chose VGG-16, ResNet, Inception-v4, and DenseNet [23] architectures with the sum, multiply, and concatenate operations on features extracted from the baseline model. They observed that element-wise sum operation on feature maps helps in better abnormality detection compared to the other two operations. Lin et al. [24] proposed a Graph Convolution Network (GCN) based self-supervising learning model known as MCG-Net. GCN is utilized to capture contextual information for multi-label fundus images whereas self-supervising learning is used for the generalization of the network. In comparison to the backbone network, their model showed performance enhancement for fundus disease classification on ODIR dataset. Ou et al. [25] proposed a two-input CNN-based attention model with a multi-scale

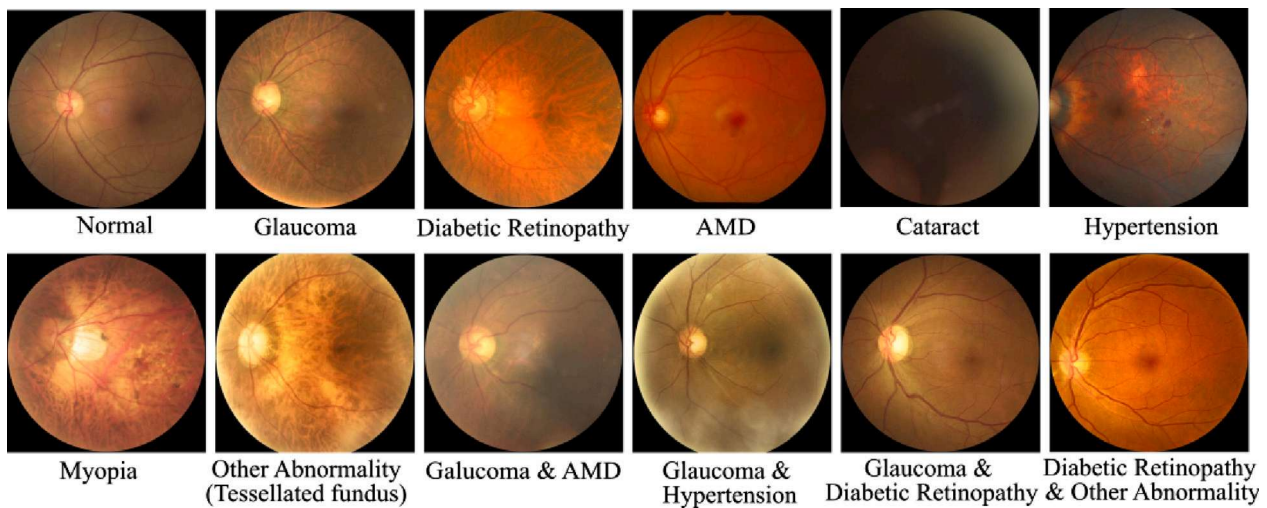


Fig. 2. Fundus images of different disease classes from ODIR-5K dataset [10].

module for multi-label fundus image classification. Multi-scale module utilized 3×3 and 1×1 dilated convolution filters to capture multi-scale features. A spatial attention module is used for feature enhancement and learning inter-dependency between global and local information. The model is found computationally efficient but could not beat the model proposed by Li et al. [19].

ODIR-5k has a common target label for a pair of fundus images. Due to the high variation in the sample counts of each class, it suffers from the class imbalance problem. None of the state-of-the-art methods attempts to address this issue. Class imbalance is a well-known issue in medical image classification, yet limited research is available on it in the context of deep learning methods [26]. Pratt et al. [27] implemented a CNN-based model for the five-class classification of diabetic retinopathy disease. The authors demonstrated that the class-weight strategy can be used to resolve over-fitting and class imbalance issues. Buda et al. [26] examined the impact of class imbalance on classification problems using CNN. In their experimentation, CIFAR-10, MNIST, and ImageNet datasets are sub-sampled to construct artificially balanced datasets. Subin et al. [28] proposed an optimized CNN-based model for disease classification. They used the flower pollination optimization algorithm in conjunction with CNN to extract the features. A Multiclass Support Vector Machine (MSVM) is used at the final layer for classification. However, they did not include minor classes for experimentation and only five disease classes of the ODIR dataset are used. Although their model exhibits good performance among the state-of-the-art methods, excluding minority classes results in the non-availability of prediction for those classes. Also, the model serves for multi-class classification but cannot support the multi-label classification of eye diseases. In CNN, the receptive field can be made large with increasing kernel size, but this usually results in an increasing number of learning parameters, which may lead to an over-fitting problem [29]. To overcome this issue, Yu et al. [29] proposed dilated convolution operation which can enlarge the receptive field without adding additional computational cost. Since large receptive fields may not be able to recognize small objects, multi-scale feature extraction can be utilized to improve image classification. Qi Zhang [30] proposed a dilated convolution-based network to extract multi-scale features with increased receptive field size. The model showed improved performance by extracting broader and deeper semantic information for liver tumour classification. Tao et al. [31] proposed a multi-scale hybrid dilated convolution module for segmentation. They used multiscale dilated convolution with variable dilation rate in the encoder-decoder architecture. Simulated on several backbones, CNN based model achieved improved performance for object segmentation.

Similarly, Sun et al. [32] proposed a self-attention-based hybrid graph convolution network with LightGBM classifier for multi-label classification of fundus images. They proposed two input models utilizing a parallel CNN-based EfficientNet for feature extraction. The extracted features are then stitched together followed by a fully connected layer to obtain the disease label. Although their model achieved the best performance among state-of-the-art methods, it is not able to identify the presence of disease in the specific fundus image. In this manuscript, a novel attention-based model is proposed with improved multi-label classification accuracy by resolving the class imbalance issue of the dataset.

4. Methodology

The proposed DKCNet model aims for multi-label classification of ophthalmic diseases. As shown in Fig. 3, It is composed of a backbone network, an attention block, and a Squeeze-and-Excitation (SE) block. Before passing to the backbone network, fundus images are pre-processed, and the class-balancing issue is addressed through oversampling and undersampling. Fundus images are pre-processed to identify the field of view. The publicly available ODIR-5K dataset designed for multi-label classification of fundus images is highly imbalanced. For class balancing, left and right eye images are separated with their corresponding ground truths and the data samples of all classes are equalized using oversampling and undersampling methods. This balanced input data is then passed to the backbone network to extract features. These features are passed through a channel shuffle operation followed by DKCNet attention block. The attention block allows the model to learn multi-scale discriminative features. These refined feature maps are then passed to the SE block, which excites channel-wise information. The features from the SE block are sent to a global average pooling layer and fully connected layer to predict the probability of disease labels. The details of these operations are explained in the following sub-sections.

4.1. ODIR-5K dataset

ODIR-5K dataset used for experimentation in this work is made available online through a grand challenge by Peking University. The dataset contains around 5000 organized fundus images of the left and right eyes of patients. These fundus images were captured using fundus cameras like Kowa, Zeiss, and Cannon, which have different image resolutions. Disease diagnostic keywords were assigned to these images by eye specialists [10]. Based on these diagnostic keywords, the disease classification labels are assigned to each pair of fundus images. This

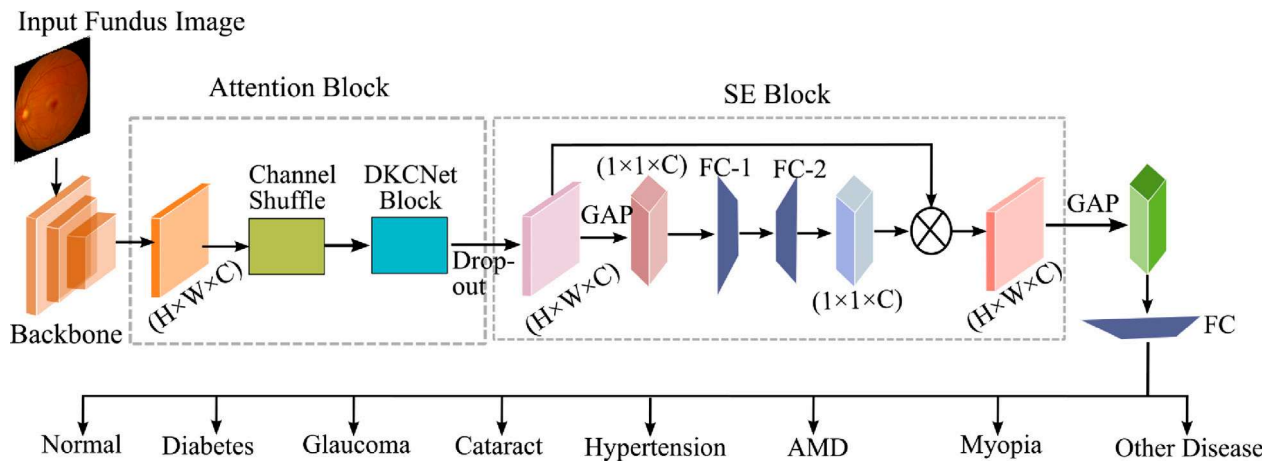


Fig. 3. Block diagram of the proposed DKCNet architecture.

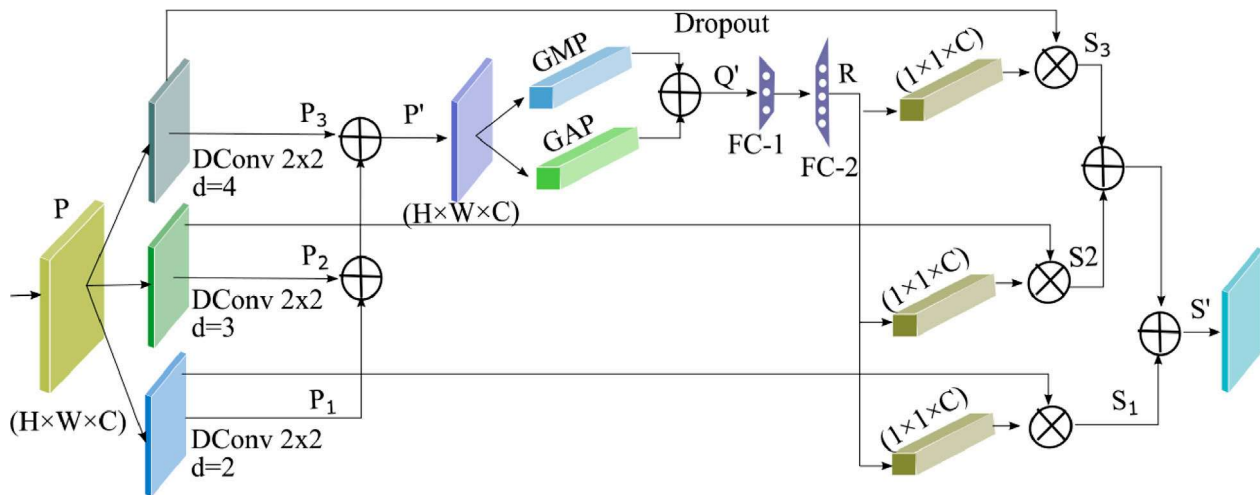


Fig. 4. Description of the DKCNet block.

visual pathology dataset is unique in comparison to other publicly accessible data sets as it contains colour fundus images of both left and right eyes of a patient with single/multiple abnormalities in a single image. The dataset contains a common target label for the pair of eye images. The images are grouped into eight disease classes, Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), AMD (A), Hypertension (H), Myopia (M), and Other diseases/abnormalities (O). The age and gender of patients are also included. Another challenging part of the ODIR-5K dataset is that the images in Other diseases class contain lesions related to 12 different ophthalmic diseases. It is not easy to learn appropriate features in such a case. Also, the dataset is highly imbalanced, considering the number of images in each of the eight classes. These issues negatively affect the accuracy and loss of the models trained for classification. Although ODIR-5K dataset is more applicable to real-life clinical situations as the images are captured with different cameras in different illumination conditions, this imposes a great challenge for any disease identification model.

4.2. Pre-processing

Most deep neural networks require the dimension of input images in a 1:1 aspect ratio. Images in the ODIR-5K dataset have different resolutions as these are captured from different cameras. Image crop operation is utilized to make these images appropriate for model training. In image crop operation, the field of view is identified by seeking the start position of the non-black pixel in the input fundus

Table 1

ODIR-5K dataset class statistics in original, after oversampling, and after undersampling operation.

Class (Label)	Samples	Oversampling		Undersampling	
		CBF	Samples	CBF	Samples
Normal (N)	1135	0	1135	12	95
Diabetes (D)	1131	0	1131	11	103
Glaucoma (G)	207	5	1035	2	104
Cataract (C)	211	5	1055	2	106
AMD (A)	171	7	1197	2	85
Hypertension (H)	94	12	1128	1	94
Myopia (M)	177	6	1062	2	86
Other diseases/abnormalities (O)	944	0	944	10	95

image. This position is used to identify the image mask border for crop operation. To support different models, the image size needs to be adjusted explicitly. Therefore, the size of cropped image is kept to 224×224 pixels as it is commonly accepted size by most of the models [21,23,33–35].

4.3. Class balancing

The left and right fundus images are treated individually as input to the CNN architecture in this work. Based on the disease description appearing for a particular eye part, the disease label is assigned to each image. A fundus image having multiple diseases is treated separately

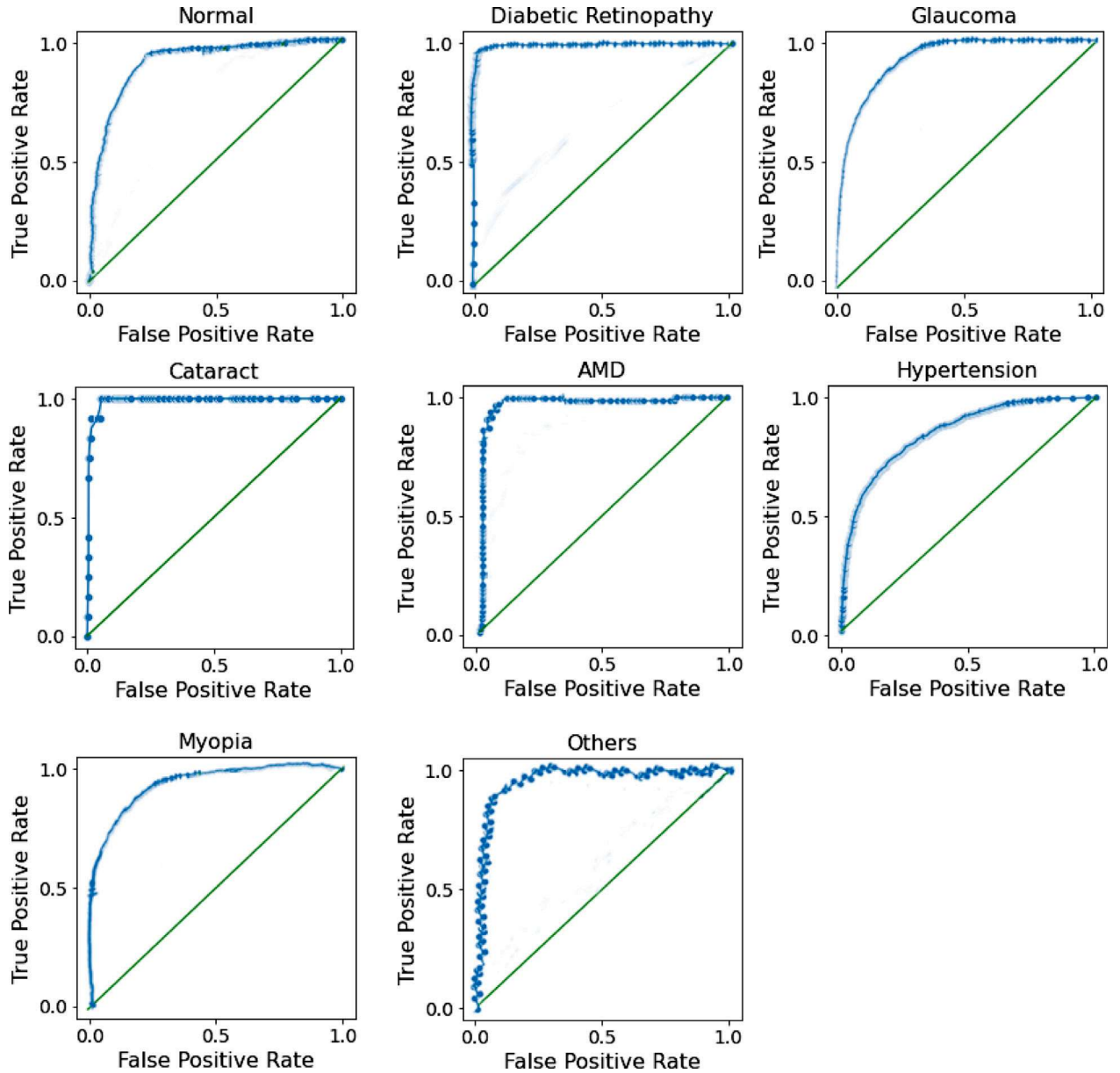


Fig. 5. AUC curves obtained with the proposed model for multi-class multi-label classification of ophthalmic diseases.

with individual disease class labels. The images with artefacts like “low-quality image”, “optical disk photographically invisible”, “lens dust”, and “image offset” are removed from the final dataset to reduce the false recognition rate. After separating left and right eye images with their corresponding ground truth, the class-wise distribution is shown in Table 1. It is observed that three classes, i.e., Normal, Diabetes, and Other diseases, have a significantly higher number of images in comparison to the remaining five classes. Random sampling techniques are used in this work to address the class balancing issue. Oversampling and undersampling are done to create two different versions of the datasets for implementation of the proposed CNN model.

4.3.1. Oversampling

The most popular approach for producing synthetic image samples is to generate images with random attributes for minority classes. New samples for Hypertension, Myopia, Cataract, AMD, and Glaucoma classes are synthetically generated using different data augmentation strategies. The new sample size is calculated as per Eq. (1).

$$M_{minor} = N_{minor} \times (1 + k) \quad (1)$$

where N_{minor} and M_{minor} represent the total number of samples in a minor class and synthetically generated samples for that class, respectively. Here, k is defined as a class balancing factor (CBF) having a value ranging from 1 to 13, which represents the number of augmentation operations. For data augmentation, flip, scaling (with ratio 0.5, 0.7, 0.8 and 0.9), crop, rotation, contrast change, hue, saturation, and gamma value change operations are used.

4.3.2. Undersampling

In this case, the whole dataset is shuffled and random images from majority classes have been selected to limit majority classes within the range of minority classes as per Eq. (2).

$$M_{major} = \left\lfloor \frac{N_{major}}{k} \right\rfloor \quad (2)$$

where N_{major} , M_{major} represent the total number of samples in a major class and random samples for that class within a range, respectively. Again, k is defined as the class balancing factor (CBF).

In this manner, image statistics for all eight disease classes get equalized.

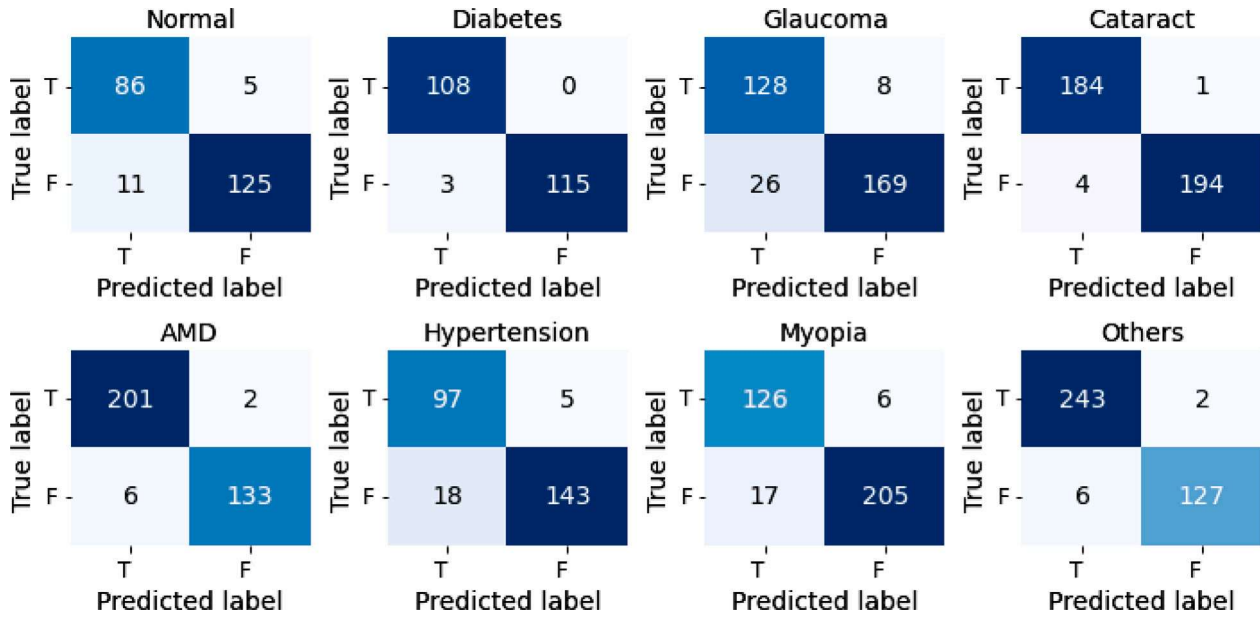


Fig. 6. Confusion matrix obtained with the proposed model for detection of various ophthalmic diseases.

4.4. The proposed DKCNet architecture

As shown in Fig. 3, the backbone network is utilized to acquire the global feature maps for fundus images. A CNN-based model pre-trained on the ImageNet dataset is used as the backbone network to extract high-level semantic features of fundus images. The output of the backbone network is fed to the attention block, which learns region-wise features to discriminate lesion parts. A channel shuffle operation is applied to the input feature maps before passing them to the DKCNet block. This permutes the channel and permits data stream across feature channels. DKCNet block (Fig. 4) utilizes dilated convolution to capture multi-scale features without adding extra computational cost. The standard convolution with a fixed kernel captures contextual information by sliding on the feature maps. In this case, features of a similar group of pixels may have a different representation in other regions, resulting in intra-class inconsistency. It is widely accepted that more contextual information can be captured by generating multi-scale features using different receptive field sizes [36,37]. Dilated convolution captures multi-scale information by varying kernel sizes known as dilation rate. With a large receptive field size, more semantic information can be captured.

The DKCNet block applies dilated convolution operation using 2×2 kernel with the dilation rate ranging from 1 to 3. Dilated convolution (DConv) is followed by batch normalization (f_{BN}) and ReLU activation (f_{ReLU}) function as shown in Eq. (3).

$$P_i = f_{ReLU}(f_{BN}(DConv_d(P))), i = \{1, 2, 3\}, d = \{2, 3, 4\} \quad (3)$$

Features obtained after dilated convolution operation are grouped by element-wise addition operation as per Eq. (4).

$$P' = P_1 \oplus P_2 \oplus P_3 \quad (4)$$

Now the spatial information is squeezed from the feature maps by performing global average pooling (f_{GAP}) and global max-pooling (f_{GMP}) operations followed by element-wise addition to get global spatial information as per Eq. (5).

$$Q = f_{GMP}(P') \oplus f_{GAP}(P') \quad (5)$$

This squeezed spatial information is passed through two fully connected layers for channel dimension reduction by a factor r . Some of the features are then dropped by introducing a feature drop layer with

a dropout rate of 0.25 followed by sigmoid (σ) activation function as per Eqs. (6) & (7).

$$Q' = f_{Drop=0.25}(Q) \quad (6)$$

$$R = f_{Sigmoid}(f_{FC}(Q')) \quad (7)$$

So obtained squeezed information vector is element-wise multiplied with feature maps obtained with dilated convolution as shown in Eq. (8).

$$S_i = R \otimes P_i, i = \{1, 2, 3\} \quad (8)$$

$$S' = S_1 \oplus S_2 \oplus S_3 \quad (9)$$

Finally, as per Eq. (9), the obtained features are added together to get the attention map. A dropout layer follows this to reduce over-fitting with a drop rate of 0.3.

These discriminative features are processed by SE block [38], which dynamically re-calibrates channels. Usually, the network gives equal weightage to all channels, SE block is used to excite informative features in different channels. The SE block squeezes the features obtained from the attention block into a vector of size C and then passes it through a two-layer neural network (FC-1 and FC-2). This neural network generates a vector containing channel-wise weights of feature maps defining the importance of each feature in each channel. Now input feature map to the SE block is multiplied by this vector to get the final attention map. Finally, a global average pooling layer followed by a fully connected layer performs disease classification by predicting class label probability.

4.5. Experimentation setup

In this study, a backbone network is selected via experimentation with pre-trained ResNet, InceptionV3, and InceptionResNet architectures. The ODIR-5K dataset is split into 80% and 20% for the training and validation set, respectively. For oversampling, the CBF value is selected as 12, 6, 5, 7, and 5 for Hypertension, Myopia, Cataract, AMD, and Glaucoma disease classes, respectively; and 0 for Normal, Diabetes, and Other diseases classes. Similarly, for under-sampling, the CBF value is selected as 12, 11, 10, and 1 for Normal, Diabetes, Other diseases, and Hypertension classes; and 2 for Glaucoma, Cataract, AMD, and

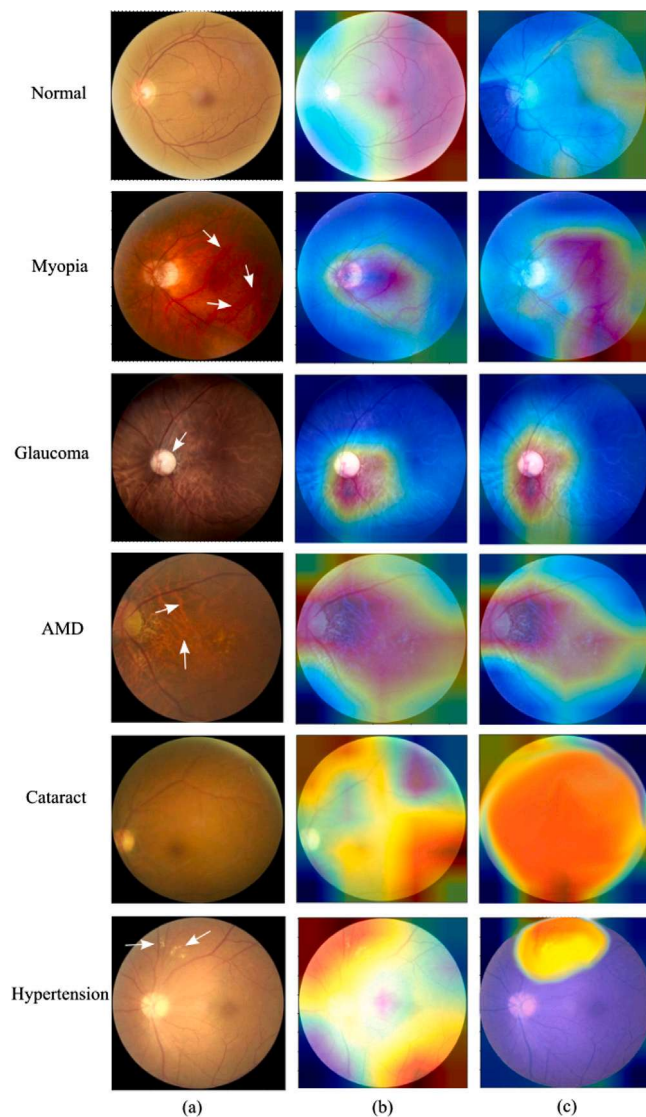


Fig. 7. Class activation maps obtained with DKCNet for single diseased fundus images from ODIR-5K dataset: (a) original fundus images with lesion parts highlighted using white arrows, (b) heatmaps generated by backbone network, and (c) heatmaps refined by DKCNet.

Myopia classes. Samples synthetically generated by both oversampling and undersampling are depicted in Table 1. The model is optimized with SGD optimizer. The initial learning rate is set to 0.0005 with decay factor $1e^{-6}$ along with the batch size as 16. All the models are trained for 100 epochs with the binary cross-entropy loss function. All experimentation work is performed on NVIDIA T4 GPU with 16 GB memory. Two experimentation scenarios have been implemented to investigate the performance of DKCNet. In the first scenario, the model is trained with a backbone network for the classification of eight ophthalmic diseases. In the second scenario, training is done on the fusion of the backbone with DKCNet.

5. Results and discussion

The performance of the proposed model is evaluated using Area Under the receiver operating Curve (AUC), F1-Score, and kappa score. The effectiveness of DKCNet is experimentally investigated, and results for 10-fold cross-validation are shown in Table 2. The proposed method can be applied to a wide range of backbone networks to improve ophthalmic disease classification. ResNet-101, Inception V3, and

InceptionResNet backbone networks are implemented with three different baselines (without sampling, oversampling and undersampling) in the proposed approach. The increase in performance can be seen in both oversampled and undersampled datasets compared to those without sampling. Results reported in Table 2 indicate that the backbone network integrated with DKCNet achieves a significant performance improvement (96.08 AUC, 94.28 F1-Score, and 0.81 kappa-score) for ophthalmic disease classification with oversampling.

To demonstrate the effectiveness of the DKCNet block, the backbone network was simulated with three popular attention methods with oversampling: Convolution Block Attention Module (CBAM), Bottleneck Attention Module (BAM), and Multi-Scale Attention Network (MSA-Net). On ODIR-5K dataset, the CBAM yielded a 94.34% AUC and 93.1 F1 score. BAM yielded 92.54% AUC and 90.42 F1 score. Similarly, MSA-Net yielded 87.02% AUC and 89.69 F1 score. The CBAM model required 94G FLOPS, the BAM model required 93G FLOPS, and MSA-Net required 192G FLOPS computations. The results given in Table 2 showcase that the discriminative kernel convolution-based attention block, i.e., DKCNet proposed in this work gives better results.

For each test image, we assigned target labels with a confidence greater than 0.5 to be positive and compared them with ground truth labels. Fig. 5 shows the AUC curves for each disease class. Similarly, from Fig. 6, it can be observed that the proposed model can identify class labels efficiently.

The proposed model is also compared with eight recent multi-class, multi-label ophthalmic disease classification models to show its efficacy. The results are reported in Table 3. As mentioned in Section 3, Islam et al. [17] considered a shallow CNN model which could not perform well for multi-label classification. Wang et al. [18] obtained a better F1-score with the EfficientNet model but lacked AUC compared to other methods. Gour and Khanna [20] achieved slightly better performance. But they utilized a heavy VGG-16 model, which does not contain a batch normalization layer, making it hard to converge. The performance obtained by Ning Li et al. [22] by using Inception-v4 model with element-wise sum feature fusion is comparable with that obtained by [20]. The spatial correlation model proposed by Li et al. [19] delivered better outcomes for different mixes of ResNet structures. The graph convolution-based model proposed by Sun et al. [32] performs better among other models in terms of F1-score and AUC. However, the proposed DKCNet model achieves improvement in AUC by 2.5% and 2.75%, respectively as compared to those achieved by Li et al. [19] and Sun et al. [32]. Table 3 also shows that the proposed network requires lesser floating-point operations per second (FLOPS) compared to state-of-the-art methods.

To check the bias of the proposed model towards training data, it is further tested on three publicly available benchmark datasets: Messidor (Diabetic Retinopathy), G1020 (Glaucoma), and Joint Shantou International Eye Centre (Multi-class dataset). From Table 4, it can be observed that the proposed DKCNet can predict retinal diseases effectively on completely unseen fundus image datasets also.

Qualitative analysis of results is performed by visualizing activation maps using Grad-CAM [39]. In Fig. 7, the first column shows fundus images containing a single disease class with lesion parts marked with white arrows. The results of the backbone network, i.e., Inception-Resnet are visualized in the second column. The third column shows refined activation maps obtained by using DKCNet with the backbone network. In the first row, the results generated by the backbone network show false detection of the lesion parts in a normal eye image, whereas the proposed model discriminates such situations effectively. Similarly, the backbone network failed to detect some lesion parts in the input image as shown in the second and sixth row, but the proposed model identifies those efficiently. The third row corresponds to the cases where the prediction results obtained with the backbone network and the proposed model are similar. In the fourth row, the backbone network falsely highlights the lesion part in the larger portion of the eye, whereas the proposed model refines the detection and is near the

Table 2

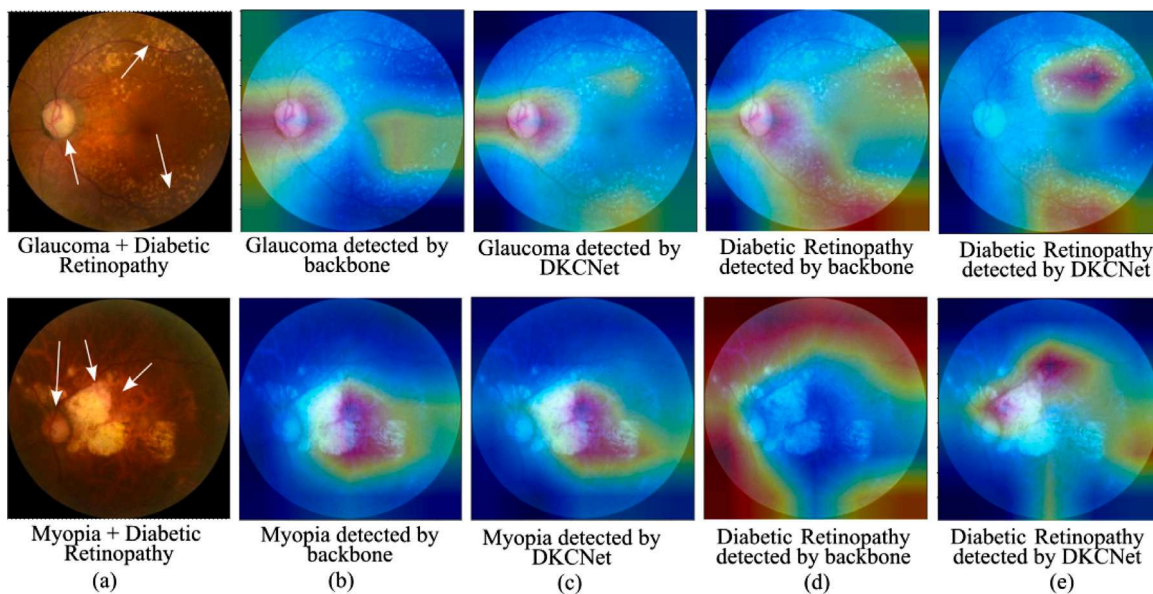
Ophthalmic disease classification by using DKCNet block with state-of-the-art backbone networks on ODIR-5K dataset.

Model	No Sampling		Undersampling		Oversampling	
	AUC	F1-Score	AUC	F1-Score	AUC	F1-Score
ResNet-101	70.12	78.53	87.04	89.56	94.33	92.28
ResNet-101 + DKC block	70.26	79.28	86.53	88.15	93.52	91.62
InceptionV3	72.94	78.17	87.45	88.93	86.31	87.37
InceptionV3 + DKC block	73.86	79.44	88.24	88.93	88.59	89
InceptionResNet	74.22	80.44	86.94	88.68	94.24	91.53
InceptionResNet + DKC block	74.55	80.96	88.05	88.93	96.08	94.28

Table 3

Ophthalmic disease classification results obtained with different methods on ODIR-5K dataset.

Author	Method	AUC	F1-Score	Params (M)	Flops (G)
Islam et al. [17], 2019	Shallow CNN	80.5	85	1.1	–
Wang et al. [18], 2020	EfficientNet	73	88	–	–
Gour and Khanna [20], 2020	Two Input VGG-16	84.93	85.57	15.2	80.2
Li et al. [19], 2020	ResNet-101	93	91.3	74.2	68.7
Ning Li et al. [22], 2021	Inception-v4	88	85.93	–	–
Lin et al. [24], 2022	Graph Conv. Network	78.16	89.66	–	–
Ou et al. [25], 2022	ResNet-50	90.3	88.6	82.6	67
Sun et al. [32], 2022	EfficientNet-B4	93.50	91.60	–	–
Proposed method	InceptionResnet + DKC Block	96.08	94.28	87.7	13.4

**Fig. 8.** Class activation maps obtained with DKCNet for multiple diseased fundus images from ODIR-5K dataset: (a) original fundus images with lesion parts highlighted using white arrows, (b) and (d) heatmaps generated by backbone network, and (c) and (e) heatmaps refined by DKCNet.**Table 4**

Cross-dataset performance evaluation of the proposed DKCNet model.

Testing dataset	AUC	F1-Score
Messidor (Diabetic Retinopathy)	89.37	87.75
G1020 (Glaucoma)	93.14	91.42
Joint Shantou International Eye Centre (Multi-class dataset)	94.18	91.15

ground truth. In the fifth row, the complete image is cloudy due to cataract disease. The backbone network highlights only a particular region, whereas the proposed model highlights the entire fundus image.

Similarly, multi-class classification performance can be visualized in Fig. 8. Here, the first column shows input fundus images having multiple diseases. The second and third columns correspond to class activation maps generated by the backbone network for the predicted class, whereas the last two columns demonstrate a refined class activation map obtained with DKCNet with better visual classification. In the fourth column of the first row of Fig. 8, it can be seen that the

backbone network failed to detect some of the lesion parts, whereas the proposed method detects those well. In the second row, the backbone network shows false detection for lesion parts, whereas the proposed DKCNet highlights those parts more accurately, as seen in the fourth and the last column of Fig. 8. For real-life applications, a GUI has been developed that takes a fundus image as an input and displays the predicted multiple disease label as shown in Fig. 9.

6. Conclusion and future work

DKCNet proposed in this work enables CNN based model to learn discriminative features with an attention module without introducing extra cost. The novelty of this work lies in a model which improves ophthalmic disease classification performance and solves the class balancing issue of the highly imbalanced ODIR-5K dataset having multiple common labels for the fundus image pair of a patient's left and right eye. DKCNet is composed of an attention block followed by a SE block. The attention block takes features from the backbone

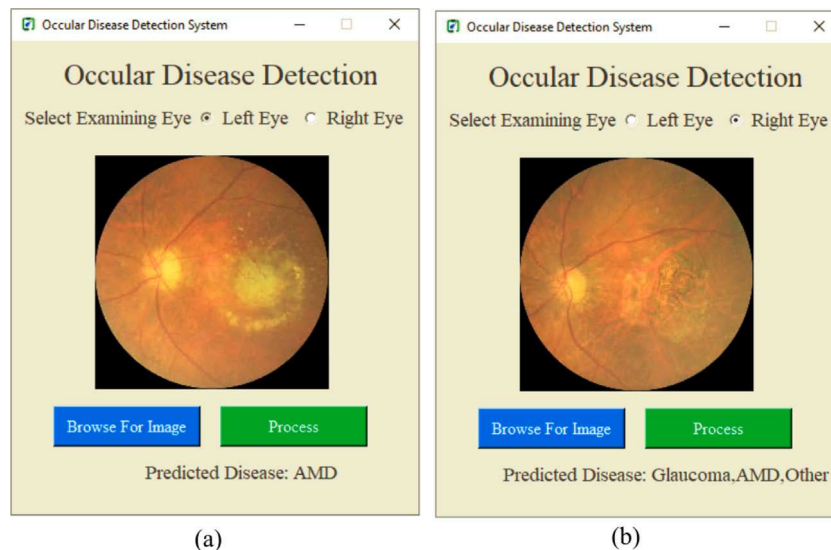


Fig. 9. Interactive GUI to predict multiple diseases in the fundus images of a patient: (a) left eye and (b) right eye.

network and generates discriminative feature attention maps. The SE block takes the discriminative feature maps and performs channel-wise attention. The experimentation has been done with three backbone CNN-based architectures, and it is found that the proposed DKCNet shows superior performance compared to counterpart methods with the InceptionResnet model.

The proposed DKCNet attention block enables multi-scale feature fusion and learns lesions of varying size and shape. Although the network does not add computational cost, it increases the redundancy of the information, which results in a higher memory footprint for the model. The future work includes addressing this memory-computational time tradeoff and reducing the memory footprint of the model. To further address the class imbalance problem, synthetic fundus image generation can be considered using methods like generative adversarial networks. The proposed method can also be extended for lesion detection and localization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] M.J. Burton, J. Ramke, A.P. Marques, R.R. Bourne, N. Congdon, I. Jones, The lancet global health commission on global eye health, in: *Vision beyond 2020*, Lancet Glob Health, 2021, pp. e489–e551.
- [2] R. Kawasaki, J. Grauslund, Clinical motivation and the needs for RIA in healthcare, in: *Computational Retinal Image Analysis*, Elsevier, 2019, pp. 5–17.
- [3] M.D. Abramoff, M.K. Garvin, M. Sonka, Retinal imaging and image analysis, *IEEE Rev. Biomed. Eng.* 3 (2010) 169–208.
- [4] T.B. Sekou, M. Hidane, J. Olivier, H. Cardot, From patch to image segmentation using fully convolutional networks—application to retinal images, 2019, arXiv preprint arXiv:1904.03892.
- [5] H. Fu, J. Cheng, Y. Xu, D.W.K. Wong, J. Liu, X. Cao, Joint optic disc and cup segmentation based on multi-label deep network and polar transformation, *IEEE Trans. Med. Imaging* 37 (7) (2018) 1597–1605.
- [6] T. Li, W. Bo, C. Hu, H. Kang, H. Liu, K. Wang, H. Fu, Applications of deep learning in fundus images: A review, *Med. Image Anal.* 69 (2021) 101971.
- [7] U. Raghavendra, H. Fujita, S.V. Bhandary, A. Gudigar, J.H. Tan, U.R. Acharya, Deep convolution neural network for accurate diagnosis of glaucoma using digital fundus images, *Inform. Sci.* 441 (2018) 41–49.
- [8] H. Fu, J. Cheng, Y. Xu, C. Zhang, D.W.K. Wong, J. Liu, X. Cao, Disc-aware ensemble network for glaucoma screening from fundus image, *IEEE Trans. Med. Imaging* 37 (11) (2018) 2493–2501.
- [9] N. Gour, M. Tanveer, P. Khanna, Challenges for ocular disease identification in the era of artificial intelligence, *Neural Comput. Appl.* (2022) 1–23.
- [10] Odir, Peking University International Competition on Ocular Disease Intelligent Recognition, ODIR-2019, 2019.
- [11] L. Lin, M. Li, Y. Huang, P. Cheng, H. Xia, K. Wang, J. Yuan, X. Tang, The SUSTech-SYSU dataset for automated exudate detection and diabetic retinopathy grading, *Sci. Data* 7 (1) (2020) 1–10.
- [12] Y. Jiang, H. Xia, Y. Xu, J. Cheng, H. Fu, L. Duan, Z. Meng, J. Liu, Optic disc and cup segmentation with blood vessel removal from fundus images for glaucoma detection, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC, IEEE, 2018, pp. 862–865.
- [13] L. Giancardo, F. Meriaudeau, T.P. Karnowski, Y. Li, S. Garg, K.W. Tobin Jr., E. Chaum, Exudate-based diabetic macular edema detection in fundus images using publicly available datasets, *Med. Image Anal.* 16 (1) (2012) 216–226.
- [14] Y. Luo, K. Chen, L. Liu, J. Liu, J. Mao, G. Ke, M. Sun, Dehaze of cataractous retinal images using an unpaired generative adversarial network, *IEEE J. Biomed. Health Inf.* 24 (12) (2020) 3374–3383.
- [15] P. Antonio, P. Marta, D. Luís, D. Antonio, S. Manuel, M. Rafael, G. Sonia, G. Manuel, M. Isabel, E. Carlos, et al., Factors associated with changes in retinal microcirculation after antihypertensive treatment, *J. Hum. Hypertens.* 28 (5) (2014) 310–315.
- [16] Z. Wang, P.A. Keane, M. Chiang, C.Y. Cheung, T.Y. Wong, D.S.W. Ting, Artificial intelligence and deep learning in ophthalmology, *Artif. Intell. Med.* (2020) 1–34.
- [17] M.T. Islam, S.A. Imran, A. Arefeen, M. Hasan, C. Shahnaz, Source and camera independent ophthalmic disease recognition from fundus image using neural network, in: 2019 IEEE International Conference on Signal Processing, Information, Communication & Systems, SPICSCON, IEEE, 2019, pp. 59–63.
- [18] J. Wang, L. Yang, Z. Huo, W. He, J. Luo, Multi-label classification of fundus images with efficientnet, *IEEE Access* 8 (2020) 212499–212508.
- [19] C. Li, J. Ye, J. He, S. Wang, Y. Qiao, L. Gu, Dense correlation network for automated multi-label ocular disease detection with paired color fundus photographs, in: 2020 IEEE 17th International Symposium on Biomedical Imaging, ISBI, IEEE, 2020, pp. 1–4.
- [20] N. Gour, P. Khanna, Multi-class multi-label ophthalmological disease detection using transfer learning based convolutional neural network, *Biomed. Signal Process. Control* 66 (2021) 102329.
- [21] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.
- [22] N. Li, T. Li, C. Hu, K. Wang, H. Kang, A benchmark of ocular disease intelligent recognition: one shot for multi-disease detection, in: *International Symposium on Benchmarking, Measuring and Optimization*, Springer, 2020, pp. 177–193.
- [23] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4700–4708.
- [24] J. Lin, Q. Cai, M. Lin, Multi-label classification of fundus images with graph convolutional network and self-supervised learning, *IEEE Signal Process. Lett.* 28 (2021) 454–458.
- [25] X. Ou, L. Gao, X. Quan, H. Zhang, J. Yang, W. Li, Bfenet: A two-stream interaction CNN method for multi-label ophthalmic diseases classification with bilateral fundus images, *Comput. Methods Programs Biomed.* (2022) 106739.
- [26] M. Buda, A. Maki, M.A. Mazurowski, A systematic study of the class imbalance problem in convolutional neural networks, *Neural Netw.* 106 (2018) 249–259.

- [27] H. Pratt, F. Coenen, D.M. Broadbent, S.P. Harding, Y. Zheng, Convolutional neural networks for diabetic retinopathy, *Procedia Comput. Sci.* 90 (2016) 200–205.
- [28] P. Muthukannan, et al., Optimized convolution neural network based multiple eye disease detection, *Comput. Biol. Med.* (2022) 105648.
- [29] F. Yu, V. Koltun, Multi-scale context aggregation by dilated convolutions, 2015, arXiv preprint [arXiv:1511.07122](https://arxiv.org/abs/1511.07122).
- [30] Q. Zhang, A novel ResNet101 model based on dense dilated convolution for image classification, *SN Appl. Sci.* 4 (1) (2022) 1–13.
- [31] T. Ku, Q. Yang, H. Zhang, Multilevel feature fusion dilated convolutional network for semantic segmentation, *Int. J. Adv. Robot. Syst.* 18 (2) (2021) 17298814211007665.
- [32] K. Sun, M. He, Y. Xu, Q. Wu, Z. He, W. Li, H. Liu, X. Pi, Multi-label classification of fundus images with graph convolutional network and lightgbm, *Comput. Biol. Med.* 149 (2022) 105909.
- [33] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 1–9.
- [34] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [35] A.G. Howard, M. Zhu, B. Chen, Kalenichenko, et al., Mobilenets: Efficient convolutional neural networks for mobile vision applications, 2017, arXiv preprint [arXiv:1704.04861](https://arxiv.org/abs/1704.04861).
- [36] H. Zhao, J. Shi, X. Qi, X. Wang, J. Jia, Pyramid scene parsing network, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 2881–2890.
- [37] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: *Proceedings of the European Conference on Computer Vision, ECCV*, 2018, pp. 801–818.
- [38] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7132–7141.
- [39] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 618–626.